FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
CONCEPT RECAPITULATION TEST – I
PAPER-2

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**C**	24.	**D**	47.	**C**
2.	**A**	25.	**B**	48.	**B**
3.	**B**	26.	**B**	49.	**C**
4.	**ABCD**	27.	**AB**	50.	**BC**
5.	**ABC**	28.	**ABCD**	51.	**ABCD**
6.	**BCD**	29.	**AC**	52.	**ABD**
7.	**ABD**	30.	**AD**	53.	**AD**
8.	**CD**	31.	**B**	54.	**AB**
9.	**D**	32.	**C**	55.	**D**
10.	**B**	33.	**B**	56.	**D**
11.	**4**	34.	**6**	57.	**6**
12.	**1**	35.	**3**	58.	**6**
13.	**3**	36.	**4**	59.	**9**
14.	**5**	37.	**3**	60.	**9**
15.	**2**	38.	**5**	61.	**2**
16.	**3**	39.	**3**	62.	**4**
17.	**1**	40.	**1**	63.	**7**
18.	**1**	41.	**8**	64.	**4**
19.	**6**	42.	**1**	65.	**1**
20.	**2**	43.	**3**	66.	**4**
21.	**00002.66**	44.	**00370.65**	67.	**00610.66**
22.	**00000.75**	45.	**00089.40**	68.	**04008.80**
23.	**00000.08**	46.	**00005.70**	69.	**08275.20**

Physics

PART – I

SECTION – A

1. C

Sol. Potential of centre of sphere = $\frac{Kq}{r}+V_i=\frac{Kq}{r}$

where V_i = potential due to induced charge at centre = 0 [$\therefore$ Σq_i = 0 and all induced charges are equidistance from centre]

$\therefore$ potential at point P = $\frac{Kq}{r}=\frac{Kq}{r_1}+V_i$ (For conductor all points are equipotential)

$\therefore$ $V_i=K\left(\frac{q}{r}-\frac{q}{r_1}\right)$

2. A

Sol. Applying Snell's law between the points O and P, we have

$2\times\sin 60^0=\left(\sin 90^0\right)\times\frac{2}{\left(1+H^2\right)}$, $2\times\frac{\sqrt{3}}{2}=1\times\frac{2}{\left(1+H^2\right)}$

$\left(1+H^2\right)=\frac{2}{\sqrt{3}}$, $H=\sqrt{\left(\frac{2}{\sqrt{3}}-1\right)}$

3. B

Sol. $\lambda=v,\frac{v}{2},\frac{v}{3}$ .....

$\lambda'=4\ell,\frac{4\ell}{3},\frac{4\ell}{5}$

$\Rightarrow$ f′ = 50, 150, 250 .....

4. ABCD

Sol. Let elongation in spring A, B and C be x_1, x_2 and x_3 respectively.
Considering spring forces and constraint relations

$x_2 = 4x_3$ …(i)

$x_2 = 2x_1$ …(ii)

and $x_1 + 2x_2 + x_3 = x$ …(iii)

$\Rightarrow$ $x_1=\left(\frac{2}{11}\right)x$; $x_2=\left(\frac{4}{11}\right)x$; $x_3=\left(\frac{1}{11}\right)x$

Also, F = $2K\left(\frac{x}{11}\right)$ $\Rightarrow$ $T=2\pi\sqrt{\frac{11m}{2k}}$

5. ABC

Sol. $\Delta U=\frac{fR\Delta T}{2}$

$\Delta W=\frac{nR\Delta T}{1-x}$ where PV^x = constant. Here $x=-\frac{1}{2}$

6. BCD

Sol. The system is equivalent to a binary system as shown in the diagram.

$r_1 = \dfrac{M \times 3R}{M + 2M} = R$

$r_2 = 2R$

Considering the circular motion of point mass m

$M\omega_2^2 r_2 = \dfrac{GM^2}{36R^2}$; $\omega_2 = \sqrt{\dfrac{GM}{72R^3}} = \omega_1$

7. ABD

Sol. For ammeter,

$i = \dfrac{i_{max}(R_s + R_A)}{(R_s)}$

$\Rightarrow$ $i = 0.1mA$ for $R_s = 50\Omega$ [as $R_A = 50\Omega$ and $i_{max} = 50\mu A$]

For voltmeter,

$V = i_{max}(R_A + R_V)$

$\Rightarrow$ $V \approx 10V$ for $R_v = 200k\Omega$

8. CD

Sol. If $\mu_1 = 0$, $\mu_2 \neq 0$

$t_1 = 0$, balancing torques about A

$N_1 \ell \sin\theta = mg\dfrac{\cos\theta}{2}$

$N_1 \tan\theta = \dfrac{mg}{2}$

If $\mu_1 \neq 0$, $\mu_2 \neq 0$

$T_2 = 0$, equilibrium cannot be attained.

If $\mu_1 \neq 0$, $\mu_2 \neq 0$

$N_1 = t_2 = \mu_2 N_2$; $N_2 + t_1 = mg$

$N_2 + \mu_1 N_1 = mg$; $N_2 + \mu_1(\mu_2 N_2) = mg$; $N_2 = \dfrac{mg}{1 + \mu_1\mu_2}$

9. D

Sol. Because resultant velocity is always perpendicular to line joining C and boat, so path is circular with center at C.

10. B

Sol. Let any time the velocity of boat with respect to river makes an angle α with *CP*.

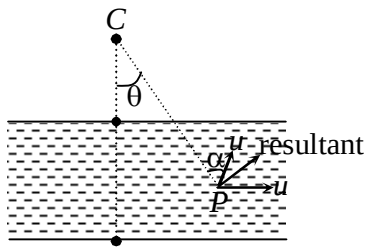
Since along *CP*, net velocity is zero

$u\cos\alpha = u\sin\theta$

$\cos\alpha = \sin\theta \Rightarrow \alpha = \dfrac{\pi}{2} - \theta$

$V_{resultant} = u\sin\alpha + u\cos\theta$

$= u\cos\theta + u\cos\theta = 2u\cos\theta$.



Angular velocity $\omega = \frac{2u\cos\theta}{2d} = \frac{u\cos\theta}{d}$, $\frac{d\theta}{dt} = \frac{u\cos\theta}{d}$

$$\int_0^{\pi/3} \frac{d\theta}{\cos\theta} = \int_0^t \frac{u}{d}dt \Rightarrow t = \frac{d}{u}\ln\left[2+\sqrt{3}\right]$$

SECTION – C

11. 4

Sol. Here 3rd maxima is shifted by 3×10^{-4} m. It indicates fringe width increases by 1×10^{-4} m.

Hence $\beta = \frac{\lambda(D+0.5)}{d} = \frac{\lambda D}{d} + 1\times10^{-4}$

or $\frac{0.5\lambda}{d} = 1\times10^{-4}$ m or $\lambda = \frac{2\times10^{-3}\times1\times10^{-4}}{0.5} = 4\times10^{-7}$ m = 400 nm

12. 1

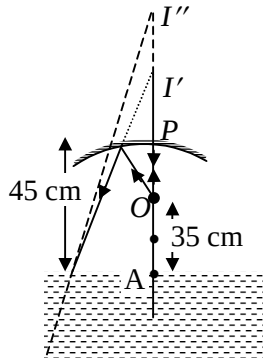
Sol. For the reflection at the concave mirror,

u = – 10 cm; v = ?; f = – 15 cm

From the mirror formula, we have

$$v = \frac{uf}{u-f} = \frac{(-10)\times(-15)}{-10+15} = \frac{150}{5} = +30 \text{ cm}$$

The positive sign indicates that the image is formed on the other side of the concave mirror,



Now, the image formed by the concave mirror serves as a virtual object for refraction at water surface which takes placed from air to water. So,

$$\mu = \frac{\text{Apparent height}}{\text{Real height}}$$

$\therefore$ AI" = Apparent height = μ × real height = $\frac{4}{3}\times75$ = 100 cm.

13. 3

Sol. Let T be the tension in the ideal string and 'a' be the acceleration of the blocks at the instant of release. For the block on the left, the upward acceleration may be found from

$T + k_1x - mg = ma$

For the block on the right, the downward acceleration may be found from

$k_2x + mg - T = ma$

Adding the equations gives the acceleration of the blocks as

$a = (k_1 + k_2)x/(2m)$

However, subtracting the equations gives

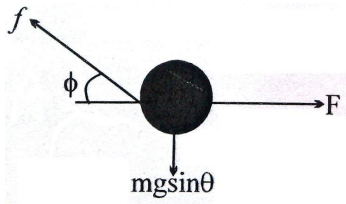
$T = mg - (k_1 - k_2)x/2$

for maximum value of k_1T will be zero.

$$mg = \left(\frac{k_1 - k_2}{2}\right)x \quad ; \quad k_1 = 300.$$

14. 5

Sol. There are four forces acting on the object; a normal force N perpendicularly away from the incline, gravity mg vertically downward, the applied force F in the direction of motion and a static frictional force f before the object begins to move. Since the object is about to slide along the plane, the frictional force must be at its maximum value, $f = \mu_s N$. The sum of the force components perpendicular to the incline is zero, so,



$N = mg\cos\theta \Rightarrow \quad f = \mu_s mg\cos\theta$ …(1)

On the other hand, the force components parallel to the surface of the incline are sketched in the following free-body diagram.

Note that the frictional force must make some angle ϕ as drawn because it initially balances the other two forces on this diagram,

$F = f\cos\theta$ …(2)

for the minimum applied force to get the object to start sliding, and

$$f\sin\phi = mg\sin\phi \quad \Rightarrow \quad f = \frac{mg\sin\theta}{\sin\phi} \quad ...(3)$$

Substitute eq. (3) into (2) along with

$$\cos\phi = \sqrt{1 - \sin^2\phi}$$

to obtain

$$F = mg\sin\theta\sqrt{\frac{1}{\sin^2\phi} - 1} \quad ...(4)$$

Also substitute eq (3) into (1) and rearrange to find

$$\frac{1}{\sin\phi} = \frac{\mu_s\cos\theta}{\sin\theta} \quad ...(5)$$

Now put eq. (5) into (4) to get

$$F = mg\sqrt{\mu_s^2\cos^2\theta - \sin^2\theta} \quad ...(6)$$

where we required $\mu_s > \tan\theta$, as otherwise the object will slip even in the absence of an applied force. Note that if $\mu_k < \mu_s$, as is usually the case, then the object will begin to accelerate after it starts moving.

15. 2

Sol. $$\frac{q_1}{C_1} = \frac{q_2}{C_2} \; ; \; q_1 + q_2 = 2Q_0$$

$$C_1 = \frac{\varepsilon_0 A}{d_0 + vt}; \; C_2 = \frac{\varepsilon_0 A}{d_0 - vt}$$

$$\frac{q_1}{q_2} = \frac{d_0 - vt}{d_0 + vt}$$

$$q_2\left(\frac{d_0 - vt}{d_0 + vt}\right) + q_2 = 2Q_0$$

$$q_2\left[\frac{2d_0}{d_0 + vt}\right] = 2Q_0$$

$$q_2 = \frac{2Q_0}{2d_0}(d_0 + vt)$$

$$I = \frac{dq_2}{dt} = \frac{Q_0 v}{d_0} = 2 \text{ amp}$$

16. 3

Sol. Initially the rod will be in equilibrium if

$2T_0 = Mg$ with $T_o = kx_o$ …(i)

when the current I is passed through the rod, it will experience a force F = BIL vertically up,

In equilibriums

2T + BIL = Mg with T = kx …(ii)

from (i) & (ii)

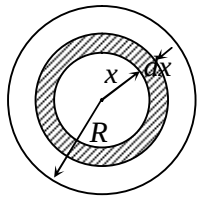
$$\frac{T}{T_o} = \frac{Mg - BIL}{Mg} \text{ i.e. } \frac{x}{x_o} = 1 - \frac{BIL}{Mg}$$

$$\text{or, } B = \frac{Mg(x_o - x)}{I \;\; L \;\; x_0}$$

Putting the values we get $B = 1.5 \times 10^{-2} T$.

17. 1

Sol. $d\varphi = (2\pi x dx)kxt^2$



$$\varphi = \frac{2}{3}kt^2\pi x^3$$

$$\frac{d\varphi}{dt} = \frac{4k\pi t x^3}{3}$$

$$E2\pi x = \frac{4\pi k t x^3}{3} \; ; \qquad E = \frac{2}{3}ktx^2 \; ; \quad d\tau = \left(\frac{2}{3}ktx^2\right)\frac{Q}{\pi R^2}(2\pi x dx)x$$

$$\int d\tau = \frac{4}{3}\frac{ktQ}{R^2}\int_0^R x^4 dx \quad \Rightarrow \quad \tau = \frac{4}{3}\frac{ktQ}{R^2}\frac{R^5}{5}$$

At t = 15sec, τ = 1 N-m

18. 1

Sol. $F = -\frac{dU}{dr} = -\frac{k}{r}, \; \frac{k}{r} = \frac{mv^2}{r}$ ..... (i)

$$E_n = \frac{1}{2}mv^2 + k \ln r \qquad \text{..... (ii)}$$

$$mvr = \frac{nh}{2\pi} \qquad \text{..... (iii)}$$

Solving these $E_n = \frac{k}{2}\left(1+\ln\left(\frac{n^2h^2}{4\pi^2mk}\right)\right)$

required ratio = $\frac{E_2-E_1}{E_4-E_2}=1.$

19. 6

Sol. $\sqrt{f_1}=\sqrt{\frac{v}{\lambda_1}}=a(11-1)$ and $\sqrt{f_2}=\sqrt{\frac{v}{\lambda_2}}=a(Z-1)$

By dividing, $\sqrt{\frac{\lambda_2}{\lambda_1}}=\frac{10}{Z-1} \Rightarrow \sqrt{\frac{4}{1}}=\frac{10}{Z-1}$

$\Rightarrow$ Z = 6.

20. 2

Sol. $\frac{dN_A}{dt}=-\lambda_1 N_A$, $\frac{dN_B}{dt}=2\lambda_1 N_A-\lambda_2 N_B$,

N_B = maximum $\Rightarrow \frac{dN_B}{dt}=0$

$\Rightarrow \quad 2\lambda_1 N_A=\lambda_2 N_{B_{max}}$

$\Rightarrow \quad N_{B_{max}}=\frac{2\lambda_1}{\lambda_2}N_A$

$\Rightarrow \quad N_{B_{max}}=\frac{2\lambda_1}{\lambda_2}N_0 e^{-\lambda_1 t}$ = 2.

SECTION – D

21. 00002.66

Sol. The observed reading will be = 10 mm + 1 × 01 mm = 1.01 cm

$\therefore$ density = $\frac{\text{Mass}}{\text{vol}} = \frac{2.7369}{(1.01)^3}$ = 2.65654 $\approx$ 2.66 gm/cm^3

22. 00000.75

Sol. $e=\frac{v_2-v_1}{u_1-u_2}$

$1=\frac{v_2-(-2)}{u_1}$

$u_1=v_2+2$

$u_1=1(-2)+5(u_1-2)$

$u_1=-2+5u_1-10$

$u_1=\frac{12}{4}=3\text{m/s}$

$v_2=1\text{ m/s}$

Kinetic energy of the given particle of mass 6 kg $= \frac{1}{2}\times(1+5)\times\left(\frac{3}{1+5}\right)^2 = \frac{3}{4}\text{J}$

23 00000.08

Sol. $D = 1\text{m}, d = 5\times10^{-3}\text{m}$

shift = $\frac{D}{d}\left[t_1(\mu_1-1)-t_2(\mu_2-1)\right]$

$= \frac{1}{5\times10^{-3}}\left[2(1.5-1)-1.5(1.4-1)\right]\times10^{-3} = \frac{0.4}{5} = 0.08\text{m}$

Chemistry

PART – II

SECTION – A

24. D

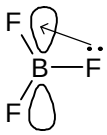
Sol. Relatively lowering will be minimum for $[Co(NH_3)_3Cl_3]$ as i = 1 and hence it will have maximum vapour pressure.

25. B

Sol. $6\overset{+4}{Xe}F_4 + 12H_2O \longrightarrow 4\overset{0}{Xe} + 2\overset{+6}{Xe}O_3 + 24HF + 3O_2$

26. B

Sol. BF_3 has back bonding



27. AB

Sol. $k = Ae^{-E_a/RT}$ or $\ell nk = \ell nA\dfrac{-E_a}{RT}$

28. ABCD

Sol. At the point of maximum value of RDF $\dfrac{dP}{dR} = 0$

$$\left(2r - \frac{2Zr^2}{a_0}\right) = 0;\ r = \frac{a_0}{Z}$$

Where Z = 3 for Li^{2+}, Z = 2 for He^+; Z = 1 for hydrogen.

29. AC

Sol. At 400 K $\Delta S_{vap} = \dfrac{\Delta H_{vap}}{T_b}$ = 100 J/mol K.

30. AD

Sol. In (A) pressure of H_2 is different
(D) conc of electrolyte is different

31. B

Sol. Antiaromatic

32. C

33. B

(Sol. for Q. no. 32 & 33)

Stronger the ligand field, more is Δ.

SECTION – C

34. 6

Sol. $P_4 + 3NaOH + 3H_2O \longrightarrow PH_3 + 3NaH_2PO_2$

X + y = 6

35. 3

Sol. $2X(g) \rightleftharpoons 2Y(g) + Z(g)$

$1 - \alpha \qquad \alpha \qquad \frac{\alpha}{2}$ Total moles = $1+\frac{\alpha}{2}=\frac{2+\alpha}{2}$

$K_P = \frac{(P_Y)^2 (P_Z)}{(P_X)^2}$

$= \frac{\left(\frac{2\alpha}{(2+\alpha)}\right)^2 P^2.\left(\frac{\alpha}{2+\alpha}\right)P}{\left(\frac{2(1-\alpha)}{(2+\alpha)}\right)^2 P^2}$ neglecting α compared to 1 and solving

$\alpha = \left(\frac{2K_P}{P}\right)^{1/3} \Rightarrow n = 3$

36. 4

Sol. Nylon-6 & Teflon are homopolymer rest all are co-polymers

37. 3

Sol. Zr, Ni & Ti are refined by vapour phase refining.

38. 5

Sol. $4H^+ + MnO_4^- \rightleftharpoons MnO_2 + 2H_2O$

$\Delta G^o = -3F\left(\frac{1.5\times 5 - 2\times 1.25}{3}\right) = -5F$

x = 5

39. 3

Sol. At 300 K volume be V

At 500 K volume = V + 2

Using $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

V = 3L

40. 1

Sol. $KMnO_4 \qquad H_2O_2$

$N_1V_1 = N_2V_2$

$5 \times 0.1 \times 10 = 28 \times N_2$

$N_2 = \frac{5}{28}$ 1 N = 5.6 V

$$\frac{5}{28}N = \frac{5.6}{10} \times \frac{5}{28} = 1V$$

41. 8

Sol. $\frac{r_+}{r_-}$ = radius ratio = $\frac{1.7}{1.8}$ = 0.944

Hence C.No. 8

42. 1

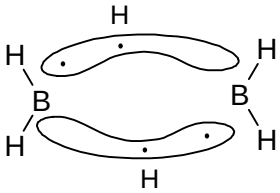
Sol. $\text{Slope} = \frac{-\Delta H}{2.303R}$

so, $\frac{-\Delta H}{2.303R} = \frac{-1}{4.606}$

ΔH = 1 cal

43. 3

Sol.



Bridge hydrogen are 3c – 2e bond

SECTION – D

44. 00370.65 K

Sol. $2.303 \log \frac{P_2}{P_1} = \frac{\Delta H}{R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$

T_2 = 100°C = 373 K P_2 = 760 mm (1 atm)

T_1 = ? P_1 = 700 mm

Δ H = 540 × 18 cal/ mol

$$2.303 \log \frac{760}{700} = \frac{540 \times 18}{2}\left(\frac{373 - T_1}{T_1 \times 373}\right)$$

T_1 = 370.65 K

45. 00089.40 kJ

Sol. $\underset{N\equiv N}{N_2} + \frac{1}{2}\underset{O=O}{O_2} \longrightarrow N = N = 0$

$\Delta H = \sum BE_R - \sum BE_P$

$= \left(946.2 + \frac{497}{2}\right) - (418 + 605.3) = 171.4\,kJ/mol$

RE = ΔH_{exp} - ΔH_{cal} = 82 – 171.4 = -89.4 kJ = 89.4 kJ

46. 00005.70

Sol. $pH = -\log K_a + \log\frac{[\text{Conj.base}]}{[\text{Acid}]}$

For - I : $4 = -\log 10^{-5} + \log\frac{[\text{Conj.base}]}{0.5}$

Conjugate base = 0.05 M = salt

For – II: $6 = -\log 10^{-5} + \log\frac{[\text{Conj.base}]}{0.5}$

[C.base] = [Salt] = 5 M

On mixing equal volumes

New conc. of salt NaA = $\frac{0.05\times V + 5\times V}{2V} = \frac{5.05}{2}M$

New conc. of HA in mixed buffer = $\frac{0.5V + 5V}{2V}$

$pH = -\log 10^{-5} + \log\frac{5.05}{2\times 0.5} = 0.5M = 5.70$

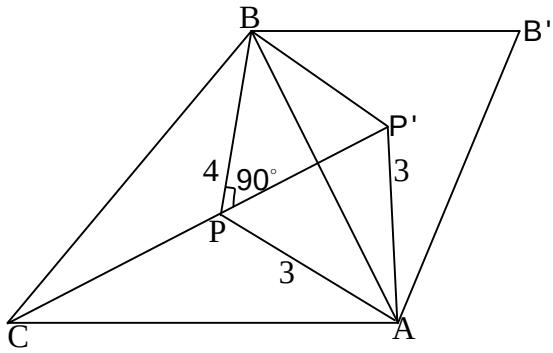
Mathematics

PART – III

SECTION – A

47. C

Sol. Rotate the triangle in clockwise direction through an angle 60°. Let the points A,B,C and P will be A,B', B and P' respectively after the rotation.
We have PA=P'A=3
And $\angle PAP' = 60^\circ \Rightarrow PP' = 3$
Also CP=BP'=5



So $\Delta BPP'$ is right angle triangle which $\angle BPP' = 90^\circ$.
Now apply cosine rule in triangle BPA because
$\angle BPA = 90^\circ + 60^\circ = 150^\circ$, PA=3 and BP=4, we can get AB

48. B

Sol. If $f'(x) > 0$ and $f''(x) > 0 \Rightarrow \dfrac{d^2 f^{-1}(x)}{dx^2} < 0$

If $f'(x) < 0$ and $f''(x) < 0 \Rightarrow \dfrac{d^2 f^{-1}(x)}{dx^2} < 0$

49. C

Sol. We have, $f(\alpha) = \displaystyle\int_{\tan^{-1}\alpha}^{\cot^{-1}\alpha} \left(\frac{\tan x}{\tan x + \cot x}\right) dx$ …(1)

$\Rightarrow \quad f(\alpha) = \displaystyle\int_{\tan^{-1}\alpha}^{\cot^{-1}\alpha} \frac{\cot x}{\cot x + \tan x} dx$ …(2)

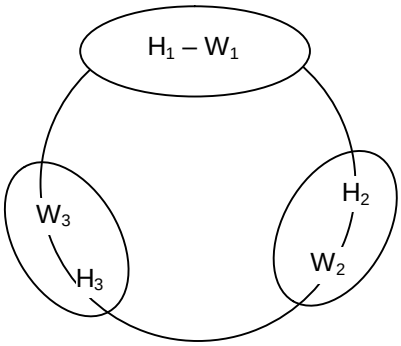
$\therefore$ Adding (1) and (2), we get

$$f(\alpha) = \frac{1}{2}\left(\cot^{-1}\alpha - \tan^{-1}\alpha\right) = \frac{1}{2}\left(\frac{\pi}{2} - 2\tan^{-1}\alpha\right) = \frac{\pi}{4} - \tan^{-1}\alpha$$

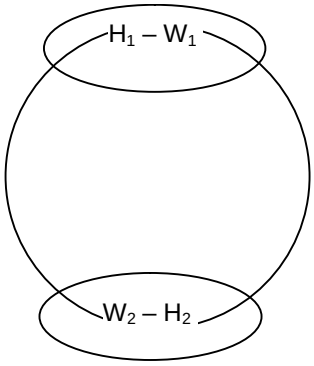
Clearly, range of $f(\alpha)$ is $\left(-\frac{\pi}{4}, \frac{3\pi}{4}\right)$.

50. BC

Sol. n (All couples are together)
$= 2 \times 2 \times 2 \times 2 \times 2 = 16$ ways
$\therefore$ x = Total – 16
$x = 5! - 16 = 105$



To find y, first Let H_1, W_1, H_2, W_2 sit in $2 \times 2 = 4$ ways



Now H_3, and W_3 can sit in 2 ways

$\therefore y = 4 \times 2 = 8$

51. ABCD

Sol. $a, b = ar, c = ar^2$ are in G.P. where $r \neq \pm 1$

$a + b + c = bx \Rightarrow 1 + r + r^2 = rx$ ……(i)

$\Rightarrow \quad r^2 + (1 - x)r + 1 = 0$

$\Delta \geq 0 \Rightarrow (1 - x)^2 - 4 \geq 0$

$(x - 1)^2 - 2^2 \geq 0, (x - 3)(x + 1) \geq 0$

$x \in (-\infty, -1] \cup [3, \infty)$

If $x = -1$, then $r = -1 \Rightarrow x = -1$ (rejected)

and $x = 3$, then $r = 1 \Rightarrow r = 3$ (rejected)

$\therefore \; x \in (-\infty, -1) \cup (3, \infty)$

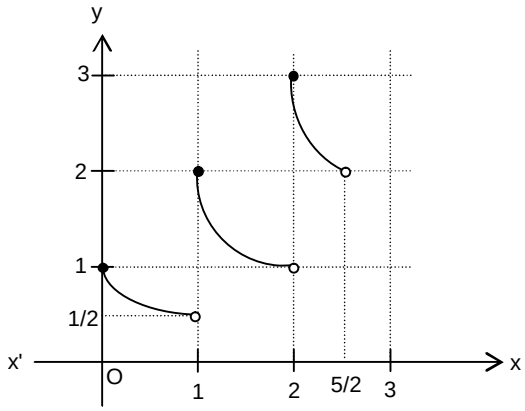
52. ABD

Sol. $$f(x) = \begin{cases} \dfrac{1}{1+x}, & 0 \leq x < 1 \\ \dfrac{2}{x}, & 1 \leq x < 2 \\ \dfrac{3}{x-1}, & 2 \leq x < \dfrac{5}{2} \end{cases}$$

Clearly, f(x) is discontinuous and bijective function

$\lim\limits_{x\to1^-} f(x)=\frac{1}{2},\ \lim\limits_{x\to1^+} f(x)=2$''

$\min\left(\lim\limits_{x\to1^-} f(x),\ \lim\limits_{x\to1^+} f(x)\right)=\frac{1}{2}\neq f(1)$



53. AD

Sol. $y=\frac{2xz}{x+z}\Rightarrow x,y$ and z are in H.P.

$\Rightarrow \frac{1}{z}-\frac{1}{y}=\frac{1}{y}-\frac{1}{x}$

$\Rightarrow \frac{x}{z}=\frac{x-y}{y-z}$

54. AB

Sol. Let E: second drawn marble is green.

$P(E)=\frac{1}{3}\left[\left\{\frac{1}{4}.1+\frac{3}{4}.\frac{1}{2}\left(\frac{1}{2}+\frac{1}{4}\right)\right\}+\frac{1}{2}\left\{\frac{2}{3}+\frac{1}{2}\left(\frac{3}{4}+\frac{1}{4}\right)\right\}+\left\{\frac{3}{4}.\frac{1}{3}+\frac{1}{4}.\frac{1}{2}\left(\frac{3}{4}+\frac{1}{2}\right)\right\}\right]$

$=\frac{1}{3}\left[\frac{73}{48}\right]=\frac{73}{144}$

n=144, m= 73 hence n-m= 71 and n-2m +2 =0

correct options are A & B

55. D

56. D

(Solution-(55 to 56)

Let $P(h,k)$ be a point with the given property and assume h, k > 0

Now, without loss of generality, we take PC > PA

So, $\sqrt{2}PC=PB+PD$

$\Rightarrow 2h^2+2(k+a)^2=(h-a)^2+(h+a)^2+2k^2+2\sqrt{(h+a)^2+k^2}\times\sqrt{(h+a)^2+k^2}$

$\Rightarrow$ locus comes out to be $x^2+y^2=a^2$

So, circumcircle of square ABCD will be locus of point P.

SECTION – C

57. 6

Sol. $\because f(x)=x^3+x^2f'(1)+xf''(2)+f'''(3)$

$\therefore\ f'(x)=3x^2+2xf'(1)+f''(2)$

Put $x=1$

$\therefore\ f'(1)+f''(2)=-3$ .....(i)

Again, $f''(x)=6x+2f'(1),\ f'''(x)=6$

Put $x=2$

$\therefore\quad f''(2)=12+2f'(1)$ .....(ii)

Solving equation (i) and (ii), we get

$f'(1)=-5$ and $f''(2)=2$

$\therefore\quad f(x)=x^3-5x^2+2x+6$

$\therefore\quad f(2)-f(1)=-6=-f(0)$

58. 6

Sol. $t_r=\dfrac{2r^2}{2r^2-60r+900}$

$=\dfrac{2r^2}{(r-30)^2+r^2}$

$\Rightarrow t_1+t_{30}=2$

But $t_{15}=1$

$\Rightarrow$ Final sum = 31

So required answer is 7

59. 9

Sol. We have $f(x)-f(-x)=6x$

$\therefore\quad f(4)-f(-4)=24$

$\Rightarrow\quad N=f(4)=24+2286=2310=2.3.5.7.11$

Hence number of divisors $=2^{n-1}=2^{5-1}=16$

60. 9

Sol. We have $\left(\overrightarrow{OP}\right)^2=\left(e^t+e^{-t}\right)^2\left(\vec{a}\right)^2+\left(e^t-e^{-t}\right)^2+2\left(e^t+e^{-t}\right)\left(e^t-e^{-t}\right)\left(\vec{a}.\vec{b}\right)$

$\left(\left(\vec{a}\right)^2=\left|\vec{a}\right|^2=9,\left(\vec{b}\right)^2=\left|\vec{b}\right|^2=16\text{ and }\vec{a}.\vec{b}=\left|\vec{a}\right|\left|\vec{b}\right|\cos\dfrac{2\pi}{3}\right)$

$\Rightarrow\left|\overrightarrow{OP}\right|^2=9\left(e^t+e^{-t}\right)^2+16\left(e^t-e^{-t}\right)^2+2\left(e^{2t}-e^{-2t}\right).3.4.\left(-\dfrac{1}{2}\right)=13e^{2t}+5e^{-2t}-14$

Now $\dfrac{d}{dt}\left|\overrightarrow{OP}\right|^2=0\Rightarrow 26e^{2t}-10e^{-2t}=0\Rightarrow e^{4t}=\dfrac{5}{13}$,

So $e^{2t}=\dfrac{\sqrt{5}}{\sqrt{13}}$

$\therefore\ \left|\overrightarrow{OP}\right|^2_{min}=13\left(\dfrac{\sqrt{5}}{\sqrt{13}}\right)+5\left(\dfrac{\sqrt{13}}{\sqrt{5}}\right)-14\Rightarrow\left|\overrightarrow{OP}\right|^2_{min}=2\sqrt{65}-14$

$\Rightarrow\ \left|\overrightarrow{OP}\right|_{min}=\sqrt{2}\sqrt{\sqrt{65}-7}=\sqrt{2}\sqrt{\sqrt{a}-b}$,

So $a=65,\ b=7$

Hence $(a+b)=72$ or $\frac{(a+b)}{8}=9$

61. 2

Sol. Given that $AA^T=4.I\Rightarrow|A|^2=4\Rightarrow|A|=\pm2$, so

$$A^T=4A^{-1}=4\frac{\text{adj}\,A}{|A|}\Rightarrow\begin{bmatrix}a_{11}&a_{21}&a_{31}\\a_{12}&a_{22}&a_{32}\\a_{13}&a_{23}&a_{33}\end{bmatrix}=\frac{4}{|A|}\begin{bmatrix}c_{11}&c_{21}&c_{31}\\c_{12}&c_{22}&c_{32}\\c_{13}&c_{23}&c_{33}\end{bmatrix}$$

Now $a_{ij}=\frac{4}{|A|}c_{ij}\Rightarrow-2c_{ij}=\frac{4}{|A|}c_{ij}\left(\text{As } a_{ij}+2c_{ij}=0\right)\Rightarrow|A|=-2$

Now $|A+4I|=\left|A+AA^T\right|=|A|\left|I+A^T\right|=-2\left|(I+A)^T\right|=-2|I+A|$

$\Rightarrow|A+4I|+2|A+I|=0$, so on comparing we get $5\lambda=2\Rightarrow\lambda=\frac{2}{5}$

Hence $5\lambda=2$

62. 4

Sol. Let $f''(x)=6a(x-1)(a>0)$ then $f'(x)=6a\left(\frac{x^2}{2}-x\right)+b=3a\left(x^2-2x\right)+b$

Now $f'(-1)=0\Rightarrow9a+b=0\Rightarrow b=-9a$

$\therefore\ f'(x)=3a\left(x^2-2x-3\right)=0\Rightarrow x=-1$ and 3

So $y=f(-1)$ and $y=f(3)$ are two horizontal tangent s $=\left|f(3)-f(-1)\right|=\left|-22-10\right|=32$

So $\frac{D}{8}$ is 4

63. 7

Sol. Given $R:r=3:1.5=2\Rightarrow\Delta ABC$ must be equilateral.

So $a=b=c=2R\sin\frac{\pi}{3}=R\sqrt{3}$ (By sine rule)

Now $a\cot^2A+b^2\cot^3B+c^3\cot^4C=R\sqrt{3}\left(\frac{1}{\sqrt{3}}\right)^2+\left(R\sqrt{3}\right)^2\left(\frac{1}{\sqrt{3}}\right)^3+\left(R\sqrt{3}\right)^3\left(\frac{1}{\sqrt{3}}\right)^4$

$=\frac{R}{\sqrt{3}}+\frac{R^2}{\sqrt{3}}+\frac{R^3}{\sqrt{3}}=\frac{3+3^2+3^3}{\sqrt{3}}=\frac{39}{\sqrt{3}}=13\sqrt{3}=m\sqrt{n}$

Hence $\left|(m-2n)\right|=7$

64. 4

Sol. Since a, b, c, d are in A.P.

$\therefore\ b-a=c-b=d-c=D$ (let common difference)

$\Rightarrow\quad d=a+3d\quad\Rightarrow\quad a-d=-3D$ and $d=b+2D\ \Rightarrow\ b-d=-2d$

Also $c=a+2D\quad\Rightarrow\quad c-a=2D$

$\therefore\quad$ Given equation $2(a-b)+4k(b-c)^2+(c-a)^3=2(a-d)+(b-d)^2+(c-d)^3$

becomes $-2D + 4kD^2 + (2D)^3 = -6D + 4D^2 - D^3$

$\Rightarrow \quad 9D^2 + (4k-4)D + 4 = 0$

Since D is real $\Rightarrow (4k-4)^2 - 4(4)(9) \geq 0$

$\Rightarrow \quad 16k^2 - 32k - 128 \geq 0 \quad \Rightarrow (4k-16)(4k+8) \geq 0$

$\therefore \quad k \in (-\infty, -2] \cup [4, \infty)$

Hence smallest positive value of k = 4

65. 1

Sol. Let $x = 10^k.a_k + 10^{k-1}.a_{k-1} + ... + 10.a_1 + a_0$ where $a_k, a_{k-1}.....a_0$ are its digits.

From $a_0.a_1.....a_k = x^2 - 10x - 22 \geq 0$

$\Rightarrow x > 11$

For every positive integer x using digits $a_k, a_{k-1}, ...a_1, a_0$

$a_0.a_1a_2.....a_k \leq a_k.9^k < a_k.10^k + a_{k-1}10^{k-1} + ... + 10.a_1 + a_0$

or, $x^2 - 10x - 22 < x$

Hence, $x < \dfrac{11+\sqrt{209}}{2} < 13$

So, $x \in (11,13) \Rightarrow$ only possible value of x is 12.

Hence answer is 1.

66. 4

Sol. $S = \dfrac{16 + 3x + 5x}{2} = 8 + 4x$

Now $3x + 5x > 16 \Rightarrow x > 2$

$3x + 16 > 5x \Rightarrow x < 8 \Rightarrow x \in (2,8)$

$5x + 16 > 3x \Rightarrow x > -8$

Now $A^2(x) = (8+4x)(4x-18)(x+8)(8-x) = (16x^2 - 64)(64 - x^2)$

Let $x^2 = t \Rightarrow t \in (4,64), f(t) = 16(t-4)(64-t) = 16(64t - t^2 - 256 + 4t)$

$\therefore f(t) = -16(t^2 - 68t + 256), f'(t) = 2t - 68 = 0 \Rightarrow t = 34$

$f''(t) = -32 < 0$

Maxima occurs at t=34

F(34)=16(30)(30)

$\therefore$ largest possible area = 120

SECTION – D

67. 00610.66

Sol. Given $y = \tan z$

$\dfrac{dy}{dx} = \sec^2 z.\dfrac{dz}{dx}$ - - - - (1)

Now $\dfrac{d^2y}{dx^2} = \sec^2 z.\dfrac{d^2z}{dx^2} + \dfrac{dz}{dx}.\dfrac{d}{dx}(\sec^2 z)$ [Using product rule]

$$= \sec^2 z.\frac{d^2z}{dx^2} + \frac{dz}{dx}.\frac{d}{dz}\left(\sec^2 z\right)\frac{dz}{dx}$$

$$\frac{d^2y}{dx^2} = \sec^2 z.\frac{d^2z}{dx^2} + \left(\frac{dz}{dx}\right)^2.2\sec^2 z.\tan z \quad \text{- - - - (2)}$$

Now $1+\frac{2(1+y)}{1+y^2}\left(\frac{dy}{dx}\right)^2$

$$= 1+\frac{2(1+\tan z)}{\sec^2 z}.\sec^4 z.\left(\frac{dz}{dx}\right)^2 = 1+2(1+\tan z).\sec^2 z.\left(\frac{dz}{dx}\right)^2$$

$$= 1+2\sec^2 z\left(\frac{dz}{dx}\right)^2 + 2\tan z.\sec^2 z\left(\frac{dz}{dx}\right)^2 \quad (3)$$

From (2) and (3), we have RHS of (2) = RHS of (3)

$$\sec^2 z.\frac{d^2z}{dx^2} = 1+2\sec^2 z\left(\frac{dz}{dx}\right)^2$$

$$\Rightarrow \quad \frac{d^2z}{dx^2} = \cos^2 z + 2\left(\frac{dz}{dx}\right)^2$$

$\Rightarrow$ $k = 2$ so $(305.33)\times k$ = 610.66

68. 04008.80

Sol. Let $A(t_1), B(t_2)$ and $C(t_3)$ be the points on the parabola

Since $AB \perp BC \Rightarrow \frac{4}{(t_1+t_2)(t_2+t_3)} = -1$

$t_2^2 + t_2(t_1+t_3) + t_1t_3 + 4 = 0$

$\Rightarrow (t_1+t_3)^2 \geq 4(t_1t_3+4) \Rightarrow (t_1-t_3)^2 \geq 16$

$\Rightarrow (t_1+t_3)^2 \geq 4(t_1t_3+4) \Rightarrow (t_1-t_3)^2 \geq 16$

also length of intercept cut off by tangents from y-axis is $|t_1-t_3| \geq 4$, hence 1002.2λ is greater than 4008.80

69. 08275.20

Sol. $(AB)(AD) = 200$

$AB + AD = 2a$

$BD^2 = AB^2 + AD^2 = (2ae)^2$

$(AB+AD)^2 = AB^2 + AD^2 + 2(AB)(AD)$

$\Rightarrow 4a^2 = 4a^2e^2 + 400 \Rightarrow b = 10$

$\Rightarrow a^2(1-e^2) = b^2 = 100 \Rightarrow a = 20$

$P = 2(AB+AD) = 4a = 80$

Or 103.44P = 8275.20

